A Diagnostic Study of Sparsity-Promoting ADMM for Multi-Obstacle Trajectory Optimization

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Abstract

Autonomous navigation requires satisfying dynamics, avoiding obstacles, and using control effort efficiently. This project studies a simplified two-dimensional double-integrator navigation problem with circular obstacles and temporally sparse control. The original goal was to use an $\ell_{2,1}$ group-sparsity penalty and an Alternating Direction Method of Multipliers (ADMM) splitting to obtain collision-free trajectories with intermittent actuation. While the ADMM formulation successfully induces sparse control, the obstacle-projection update fails to reliably transfer safety from the auxiliary variable to the dynamic trajectory.

The proposed solver separates the problem into a dynamic trajectory update, a geometric obstacle projection, and a block soft-thresholding update for group sparsity. To evaluate the method rigorously, I report safety separately for X and Z, implement adaptive penalty scaling, add an obstacle-aware linearized baseline, and run ablations over λ, ADMM iteration budget, and randomized environments. In the fixed environment, the adaptive ADMM method reduces active control timesteps from 100.0% to 40.0% and lowers the control-effort proxy $\sum_t \|u_t\|_2$ from 76.32 to 67.98, but the dynamically feasible trajectory remains unsafe with minimum clearance -1.483. In 10 randomized trials, the proposed method achieves $59.2\% \pm 2.1\%$ sparsity and reduces the average effort proxy to 85.39 ± 7.66, but the dynamic trajectory X is collision-free in $0/10$ trials while the auxiliary projection Z is collision-free in $8/10$ trials. Ultimately, the method promotes sparsity but fails to guarantee collision-free trajectories.

Introduction

A robot's trajectory must satisfy system dynamics, reach a desired goal, avoid obstacles, and avoid unnecessary control effort. Combining these objectives is challenging because obstacle avoidance is inherently non-convex.

This project studies whether structured sparsity can encourage intermittent actuation in a simple navigation model. I apply an $\ell_{2,1}$ penalty to the control sequence. Each group is the two-dimensional acceleration vector at one timestep, so the penalty can set both acceleration components to zero together. In the double-integrator model, these zero-input intervals correspond to the robot coasting.

The optimization problem minimizes a quadratic tracking objective subject to linear dynamics and circular obstacle constraints. I used ADMM splitting to separate the dynamically feasible variables (X, U) from auxiliary variables (Z, V). The Z update projects positions outside obstacles, and the V update applies block soft-thresholding to promote temporally sparse control.

The experiments reveal a systematic limitation in how ADMM handles non-convex obstacle constraints: the projected trajectory Z can be safe while the dynamically feasible trajectory X remains unsafe. This distinction matters because only X satisfies the robot dynamics. Measuring collisions using only the auxiliary variable Z obscures the actual safety of the system.

The contributions of this report are:

- A formulation of an $\ell_{2,1}$ sparsity-promoting trajectory optimization problem for a 2D double-integrator model;
- An ADMM implementation with over-relaxation, block soft-thresholding, and adaptive penalty scaling;

- An obstacle-aware linearized penalty baseline in addition to obstacle-blind quadratic and ℓ_1 baselines;
- Diagnostic experiments that separately measure sparsity, control effort, X-trajectory safety, Z-trajectory safety, and the $X - Z$ agreement gap;
- An analysis demonstrating that greedy projection cannot guarantee collision-free dynamics.

Related Work

Trajectory optimization and Model Predictive Control (MPC).
Trajectory optimization and MPC are widely used for robotic motion planning and control. MPC repeatedly solves a finite-horizon optimal control problem from the current state (Mayne et al. 2000). In legged robotics, trajectory optimization and convex MPC have been used for gait generation and real-time locomotion control (Winkler et al. 2018; Di Carlo et al. 2018). Standard quadratic penalties shrink control magnitudes, whereas I aim to produce exactly zero control intervals.

Sparse control.
Sparse control methods encourage controllers that use fewer active inputs or fewer active time intervals. Hands-off control uses sparse optimization to maximize intervals where the control input is zero (Nagahara, Quevedo, and Ostergaard 2016). Sparse feedback design has also been studied with ADMM-based optimization (Lin, Fardad, and Jovanović 2013). I aimed to enforce group sparsity across both acceleration components, rather than elementwise sparsity.

ADMM for trajectory optimization.
ADMM can be used to split the objective into simpler subproblems (Boyd et al. 2011). In robotics, ADMM-based methods have been used for multi-robot trajectory optimization and distributed nonlinear motion planning (Ni, Pan, and Gao 2022; Ferranti et al. 2023). Recent work also studies ADMM in graphs of convex set trajectory optimization (Pries et al. 2026). I aimed to apply ADMM to separate the dynamics, obstacle avoidance, and group sparsity constraints.

Problem Formulation

The robot is modeled as a discrete-time linear system

$$x_{t+1} = Ax_t + Bu_t, \qquad t = 0, \dots, T-1, \tag{1}$$

where $x_t \in \mathbb{R}^{n_x}$ is the state and $u_t \in \mathbb{R}^{n_u}$ is the control. In the experiments, I use a two-dimensional double integrator with

$$x_t = [p_{x,t}, p_{y,t}, v_{x,t}, v_{y,t}]^\top, \qquad u_t = [a_{x,t}, a_{y,t}]^\top.$$

With timestep Δt, the dynamics matrices are

$$A = \begin{bmatrix} 1 & 0 & \Delta t & 0 \\ 0 & 1 & 0 & \Delta t \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \qquad B = \begin{bmatrix} \frac{1}{2}\Delta t^2 & 0 \\ 0 & \frac{1}{2}\Delta t^2 \\ \Delta t & 0 \\ 0 & \Delta t \end{bmatrix}. \tag{2}$$

For circular obstacles with centers $c_j \in \mathbb{R}^2$ and radii r_j, a position $p_t = [p_{x,t}, p_{y,t}]^\top$ is safe when

$$\|p_t - c_j\|_2 \geq r_j, \qquad \forall j. \tag{3}$$

The signed clearance of a trajectory is

$$d_{\min}(X) = \min_{t,j} \left(\|p_t - c_j\|_2 - r_j \right). \tag{4}$$

A positive value means that all sampled positions lie outside all obstacles, while a negative value indicates collision.

Let $U = [u_0, u_1, \ldots, u_{T-1}]$. The group-sparsity penalty is

$$\|U\|_{2,1} = \sum_{t=0}^{T-1} \|u_t\|_2. \tag{5}$$

The intended optimization problem is

$$\begin{aligned} \min_{X,U} \quad & \sum_{t=1}^{T} \frac{1}{2}\|x_t - x_{\text{goal}}\|_2^2 + \lambda \sum_{t=0}^{T-1} \|u_t\|_2 + \sum_{t=1}^{T} \mathcal{I}_{\text{safe}}(x_t) \\ \text{s.t.} \quad & x_{t+1} = Ax_t + Bu_t, \qquad t = 0, \ldots, T-1, \\ & x_0 = x_{\text{init}}, \end{aligned} \tag{6}$$

where $\mathcal{I}_{\text{safe}}(x_t)$ is 0 if (3) holds and ∞ otherwise.

ADMM Method

Two-Block Splitting

The method introduces auxiliary variables Z for obstacle projection and V for sparse controls:

$$X = Z, \qquad U = V.$$

The variables are grouped into two ADMM blocks:

$$W_1 = (X, U), \qquad W_2 = (Z, V).$$

This grouping is important because direct multi-block ADMM can diverge even for some convex problems (Chen et al. 2016). In the obstacle-free convex case, the grouping restores the standard two-block ADMM structure. In the full obstacle problem, the projection step is non-convex, so I implemented this grouping to improve numerical stability, though it lacks a formal global convergence guarantee.

The scaled augmented Lagrangian is

$$\begin{aligned} \mathcal{L}_\rho = & \sum_{t=1}^{T} \frac{1}{2}\|x_t - x_{\text{goal}}\|_2^2 + \sum_{t=1}^{T} \mathcal{I}_{\text{safe}}(z_t) + \lambda \sum_{t=0}^{T-1} \|v_t\|_2 \\ & + \frac{\rho}{2} \sum_{t=1}^{T} \|x_t - z_t + y_{x,t}\|_2^2 + \frac{\rho}{2} \sum_{t=0}^{T-1} \|u_t - v_t + y_{u,t}\|_2^2. \end{aligned} \tag{7}$$

Dynamic Update

The (X, U) update solves a convex quadratic program:

$$\begin{aligned} (X^{k+1}, U^{k+1}) = \arg\min_{X,U} \quad & \sum_{t=1}^{T} \frac{1}{2}\|x_t - x_{\text{goal}}\|_2^2 + \frac{\rho^k}{2} \sum_{t=1}^{T} \|x_t - z_t^k + y_{x,t}^k\|_2^2 \\ & + \frac{\rho^k}{2} \sum_{t=0}^{T-1} \|u_t - v_t^k + y_{u,t}^k\|_2^2 \\ \text{s.t.} \quad & x_{t+1} = Ax_t + Bu_t, \qquad x_0 = x_{\text{init}}. \end{aligned} \tag{8}$$

This step enforces the system dynamics but ignores obstacles.

Over-Relaxation

I use over-relaxed variables

$$\hat{X}^{k+1} = \alpha X^{k+1} + (1-\alpha)Z^k, \qquad \hat{U}^{k+1} = \alpha U^{k+1} + (1-\alpha)V^k, \tag{9}$$

with $\alpha = 1.6$. These are passed to the projection and soft-thresholding updates. $\alpha = 1.6$ is chosen because values in $[1.5, 1.8]$ empirically accelerate convergence in many practical problems(Boyd et al. 2011).

Obstacle Projection

The obstacle avoidance constraint $I_{\text{safe}}(x_t)$ is an indicator function that evaluates to 0 if the position is outside all obstacles and ∞ otherwise. Incorporating this constraint into the Block 1 quadratic program update is intractable because the obstacle free set is non-convex, making the joint problem NP-hard. Instead, the ADMM splitting introduces an auxiliary variable Z that absorbs the obstacle constraint, reducing obstacle avoidance to a geometric projection with a cheap closed-form solution.

For a single circular obstacle with center c and radius r, a point p inside the obstacle is projected radially to the boundary:

$$\Pi_{\text{safe}}(p) = c + (r+\epsilon)\frac{p-c}{\|p-c\|_2}. \tag{10}$$

For multiple obstacles, this operation is applied sequentially. This is a greedy projection, not the exact Euclidean projection onto the complement of the union of obstacles. As a result, the consensus constraint $X = Z$ and the dual variable Y_x must carry all obstacle information back to the dynamic trajectory X. If this is not satisfied exactly after a finite number of iterations, Z can be obstacle-free while X still intersects an obstacle. This gap between X and Z safety is the central diagnostic question of this paper.

Group Soft-Thresholding

The V update applies the proximal operator of the $\ell_{2,1}$ norm, which has a closed-form block soft-thresholding solution (Boyd et al. 2011):

$$v_t^{k+1} = \left(1 - \frac{\lambda}{\rho^k \|q_t\|_2}\right)_+ q_t, \qquad q_t = \hat{u}_t^{k+1} + y_{u,t}^k, \tag{11}$$

where $(a)_+ = \max(a, 0)$. When $\|q_t\|_2 \leq \lambda/\rho^k$, the entire vector v_t is set to zero, producing a zero-input interval at timestep t.

Adaptive Penalty Scaling

Following the residual balancing heuristic in (Boyd et al. 2011), I update ρ at each iteration based on the primal residual $r^k = \hat{X}^{k+1} - Z^{k+1}$ and dual residual $s^k = \rho(Z^{k+1} - Z^k)$:

$$\rho^{k+1} = \begin{cases} \tau\rho^k, & \|r^k\|_2 > \mu\|s^k\|_2, \\ \rho^k/\tau, & \|s^k\|_2 > \mu\|r^k\|_2, \\ \rho^k, & \text{otherwise}, \end{cases} \tag{12}$$

with $\mu = 10$ and $\tau = 2$. When ρ changes, the scaled dual variables are rescaled to remain consistent with the augmented Lagrangian. The idea is to increase ρ when the primal residual is large relative to the dual residual, tightening the consensus constraint, and decrease it otherwise.

Algorithm 1 Diagnostic Two-Block ADMM for Sparse Trajectory Optimization

Initialize $X^0, U^0, Z^0, V^0, Y_x^0, Y_u^0$, $\rho^0 = 1$, $\lambda = 2.5$, $\alpha = 1.6$
for $k = 0, 1, 2, \ldots$ **do**
    Solve the dynamic QP for (X^{k+1}, U^{k+1})
    $\hat{X}^{k+1} \leftarrow \alpha X^{k+1} + (1-\alpha) Z^k$
    $\hat{U}^{k+1} \leftarrow \alpha U^{k+1} + (1-\alpha) V^k$
    Project positions in $\hat{X}^{k+1} + Y_x^k$ sequentially outside obstacles to obtain Z^{k+1}
    Apply block soft-thresholding to $\hat{U}^{k+1} + Y_u^k$ to obtain V^{k+1}
    $Y_x^{k+1} \leftarrow Y_x^k + \hat{X}^{k+1} - Z^{k+1}$
    $Y_u^{k+1} \leftarrow Y_u^k + \hat{U}^{k+1} - V^{k+1}$
    Compute residuals, X clearance, Z clearance, and $\|X - Z\|$
    Update ρ using residual balancing
end for

Experimental Design

Implementation Details

All experiments were implemented in Python with NumPy and CVXPY. The QP subproblems were solved with OSQP. The timestep was $\Delta t = 0.2$, with state dimension $n_x = 4$ and control dimension $n_u = 2$. A timestep is counted as inactive if $\|u_t\|_2 < 10^{-4}$. The reported control-effort proxy is $\sum_t \|u_t\|_2$. This is not a physical energy measurement because the model does not include actuator dynamics, motor efficiency, contact, or battery behavior.

The fixed environment has horizon $T = 40$, initial state $[0, 0, 0, 0]^\top$, goal state $[10, 10, 0, 0]^\top$, and three circular obstacles with centers $(3, 4)$, $(6, 6)$, and $(8, 2.5)$ and radii 1.5, 1.8, and 1.2.

Baselines

I compared four methods:

- **Obstacle-blind quadratic MPC:** dynamics-constrained tracking with a quadratic control penalty.
- **Obstacle-blind elementwise ℓ_1 MPC:** dynamics-constrained tracking with an elementwise ℓ_1 control penalty.
- **Linearized obstacle-penalty ℓ_1 MPC:** a convex baseline that penalizes violations of halfspace approximations to obstacle constraints.
- **ADMM $\ell_{2,1}$ sparse control:** the proposed method with fixed or adaptive ρ.

The first two baselines are useful for measuring sparsity and effort, but they are not valid safe-planning baselines because they do not include obstacle constraints. The linearized obstacle-penalty baseline is included to give a more relevant obstacle-aware comparison.

Evaluation Metrics

Evaluation is based on: active control percentage; sparsity percentage; effort proxy $\sum_t \|u_t\|_2$; peak control magnitude; minimum signed clearance of X; minimum signed clearance of Z; whether X and Z are collision-free; agreement gap $\|X - Z\|$.

Safety must be evaluated on X because only X satisfies the system dynamics.

Method	Active (%)	Sparse (%)	Effort	Peak	$d_{\min}(X)$	X Safe
Obstacle-blind quadratic MPC	100.0	0.0	76.32	23.30	-1.757	No
Obstacle-blind elementwise ℓ_1 MPC	40.0	60.0	71.48	36.37	-1.258	No
Linearized obstacle-penalty ℓ_1 MPC	35.0	65.0	158.64	64.26	0.458	Yes
ADMM $\ell_{2,1}$, fixed ρ	40.0	60.0	67.98	34.79	-1.483	No
ADMM $\ell_{2,1}$, adaptive ρ	40.0	60.0	67.98	34.79	-1.483	No

Table 1: Fixed-environment comparison. The proposed ADMM method improves sparsity and effort relative to the obstacle-blind quadratic baseline, but it does not produce a collision-free dynamically feasible trajectory X. The linearized obstacle-penalty baseline achieves safety in this fixed case but requires much larger effort and peak control.

Results

Fixed Environment Comparison

Table 1 compares the baselines and the proposed ADMM method in the fixed obstacle environment. The obstacle-blind quadratic MPC baseline uses control at every timestep, with 100.0% active control and 0.0% sparsity. It has an effort proxy of 76.32, but it is not collision-free, with $d_{\min}(X) = -1.757$. This is expected because this baseline does not model obstacles.

Adding elementwise ℓ_1 regularization makes the control much sparser. The elementwise ℓ_1 MPC baseline uses active control at only 40.0% of timesteps and reaches 60.0% sparsity. Its effort proxy also decreases slightly to 71.48. However, this baseline still collides with obstacles, with $d_{\min}(X) = -1.258$. This shows that sparsity alone does not solve the safety problem.

The linearized obstacle-penalty ℓ_1 MPC baseline's minimum clearance is positive, with $d_{\min}(X) = 0.458$. The cost of this safety improvement is much larger control usage: its effort proxy is 158.64, and its peak control magnitude is 64.26, both much higher than the obstacle-blind baselines.

The proposed ADMM method achieves the same sparsity level as the elementwise ℓ_1 baseline in this fixed environment. Both the fixed-ρ and adaptive-ρ ADMM variants have 40.0% active control, 60.0% sparsity, and an effort proxy of 67.98. This is lower than the quadratic baseline's effort of 76.32 and slightly lower than the elementwise ℓ_1 baseline's effort of 71.48. However, both ADMM approaches remain unsafe, with $d_{\min}(X) = -1.483$.

This fixed-environment result shows the main tradeoff of the method. ADMM successfully promotes sparse control and lowers the effort proxy, but it does not guarantee obstacle avoidance for the actual dynamic trajectory X. In the trajectory diagnostics, the auxiliary trajectory Z is projected outside the obstacles, but X still passes through an obstacle. The later iteration-budget results also show that the final $\|X - Z\|$ gap stays around 3.1, so the safety of Z does not transfer to the dynamically feasible trajectory X.

Lambda Sweep

Table 2 shows how the sparsity parameter λ affects the adaptive ADMM method. As λ increases, the control becomes sparser. At $\lambda = 0.10$, the method has 72.5% active control and 27.5% sparsity. At $\lambda = 5.00$, active control drops to 27.5%, and sparsity increases to 72.5%. The effort proxy follows the same general trend, decreasing from 120.20 at $\lambda = 0.10$ to 53.07 at $\lambda = 5.00$.

However, the safety result does not improve with larger λ. Every tested value of λ still has negative minimum clearance. The best clearance in the table is only $d_{\min}(X) = -1.258$ at $\lambda =$

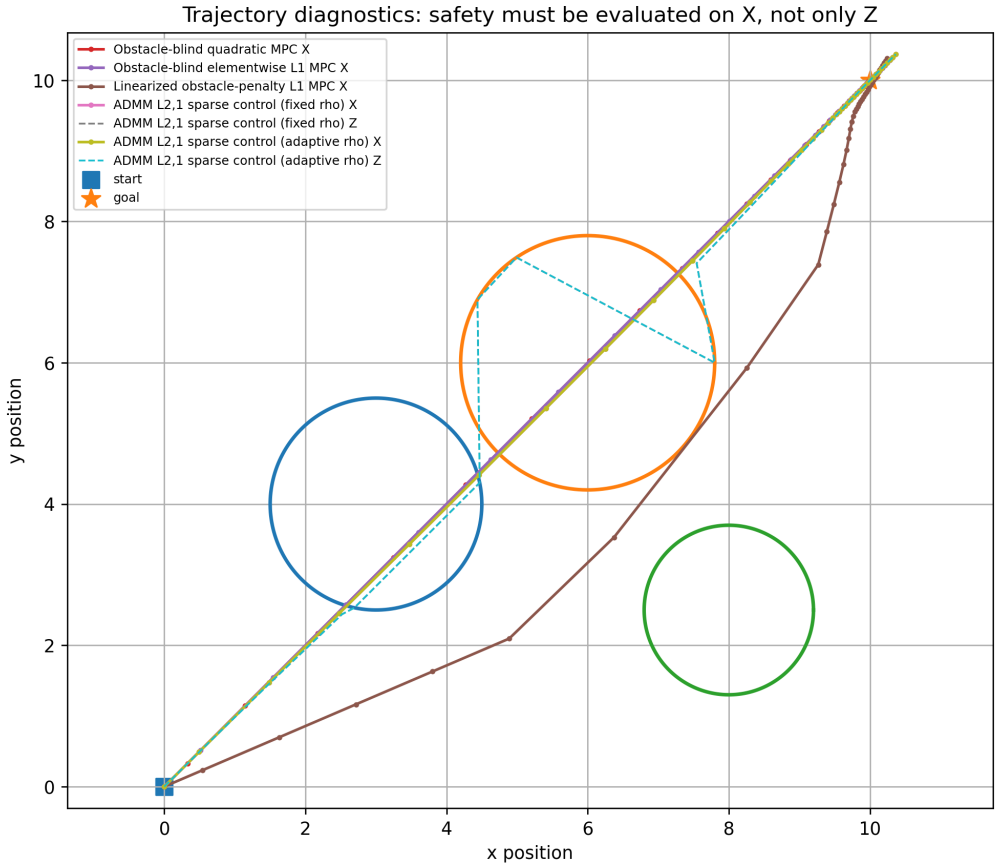


Figure 1: Fixed-environment trajectories. We compare the dynamically feasible trajectory X against the auxiliary trajectory Z. The system is only safe if X avoids obstacles.

3.91, which is still a collision. Several larger λ values actually make the clearance worse again, such as $d_{\min}(X) = -1.709$ at $\lambda = 4.46$ and $d_{\min}(X) = -1.645$ at $\lambda = 5.00$.

The $X - Z$ gap also remains large throughout the sweep. It ranges from about 2.88 to 3.33, rather than shrinking toward zero. This suggests that changing λ mainly adjusts the sparsity-effort tradeoff, but it does not fix the deeper problem that the projected auxiliary trajectory Z and the dynamically feasible trajectory X do not agree closely enough near obstacles.

Iteration-Budget Sweep

Table 3 tests whether the ADMM safety failure is mainly caused by stopping too early. I increased the maximum number of ADMM iterations from 50 to 1000. If early stopping were the main problem, then the minimum clearance of X should improve as the iteration budget increases. That is not what happens.

Across all iteration budgets, the minimum clearance stays almost unchanged. At 50 iterations, $d_{\min}(X) = -1.476$. At 120, 300, and 600 iterations, the value remains around -1.483. Even at 1000 iterations, the clearance is still negative at -1.464. Therefore, the dynamic trajectory is still colliding even after many more iterations.

The sparsity and effort values are also nearly unchanged. From 50 to 600 iterations, the method stays at 40.0% active control and about 60.0% sparsity. The effort proxy stays close to 68. The

λ	Active (%)	Sparse (%)	Effort	$d_{\min}(X)$	$\|X - Z\|$
0.10	72.5	27.5	120.20	-1.621	2.883
0.64	55.0	45.0	94.01	-1.299	2.878
1.19	47.5	52.5	83.20	-1.675	3.226
1.73	45.0	55.0	75.61	-1.501	2.921
2.28	40.0	60.0	70.10	-1.332	2.932
2.82	37.5	62.5	65.34	-1.687	3.102
3.37	35.0	65.0	61.61	-1.539	3.143
3.91	30.0	70.0	58.36	-1.298	3.082
4.46	30.0	70.0	54.96	-1.709	3.320
5.00	27.5	72.5	53.07	-1.645	3.325

Table 2: Lambda sweep for adaptive ADMM. Larger λ increases sparsity and decreases the effort proxy, but none of the tested values make the dynamically feasible trajectory X collision-free.

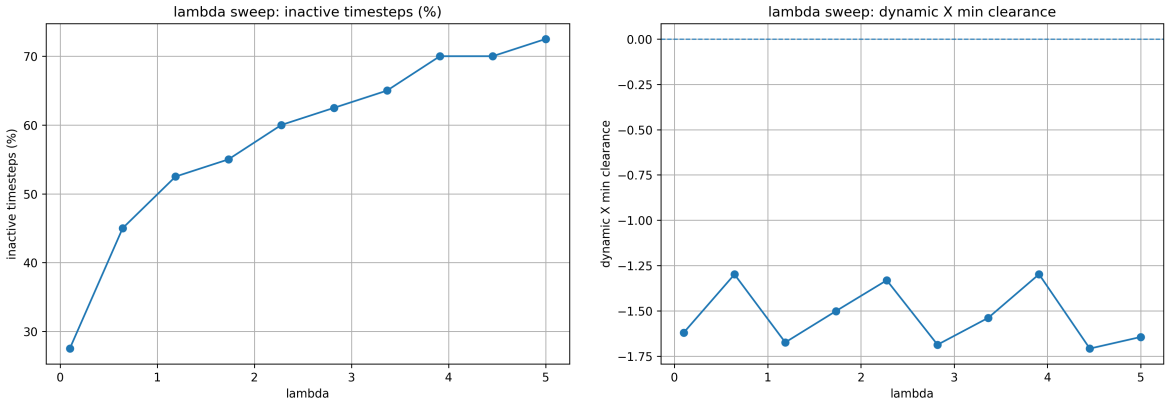


Figure 2: Lambda-sweep diagnostics. The left plot shows that λ controls temporal sparsity. The right plot shows that dynamic safety does not improve into the collision-free regime.

$X - Z$ gap also remains large, around 3.1 for most settings. These results suggest that the failure is not just a convergence-speed issue. Instead, the ADMM splitting appears to settle into a solution where Z is corrected by projection, but X remains dynamically feasible and unsafe.

Monte Carlo Validation

To check whether the same behavior appears outside the fixed environment, I tested 10 randomized environments with seed 42. Each trial sampled a horizon between 30 and 49 timesteps, an initial position in $[0, 3] \times [0, 3]$, a goal position in $[12, 15] \times [12, 15]$, and 3 to 5 circular obstacles with centers in $[4, 11] \times [4, 11]$ and radii in $[1.0, 2.5]$.

Table 4 shows the aggregate results. The obstacle-blind quadratic MPC baseline has no sparse control, with $0.0\% \pm 0.0\%$ sparsity, and it is unsafe in all 10 trials. Its average effort proxy is 91.43 ± 6.25.

The linearized obstacle-penalty ℓ_1 MPC baseline performs better on safety. It reaches $53.6\% \pm 4.6\%$ sparsity and is collision-free in $4/10$ trials. However, it also has the highest average effort, 109.23 ± 27.74. This matches the fixed-environment result: the obstacle-aware baseline can improve safety, but it often requires more aggressive control.

The adaptive ADMM method has the best sparsity-effort profile in the Monte Carlo experiments. It reaches $59.2\% \pm 2.1\%$ sparsity and has the lowest average effort proxy, 85.39 ± 7.66. However, it

Max Iterations	Active (%)	Sparse (%)	Effort	$d_{\min}(X)$	$\|X - Z\|$
50	40.0	60.0	68.22	-1.476	2.867
120	40.0	60.0	67.98	-1.483	3.097
300	40.0	60.0	67.98	-1.483	3.100
600	40.0	60.0	67.98	-1.483	3.094
1000	37.5	62.5	68.28	-1.464	3.098

Table 3: Iteration-budget sweep. More iterations do not fix the safety failure in the fixed environment, suggesting that the limitation is structural rather than only a premature-stopping issue.

Method	Sparsity (%)	Effort	X Safe	Z Safe
Obstacle-blind quadratic MPC	0.0 ± 0.0	91.43 ± 6.25	0/10	–
Linearized obstacle-penalty ℓ_1 MPC	53.6 ± 4.6	109.23 ± 27.74	4/10	–
ADMM $\ell_{2,1}$ adaptive ρ	59.2 ± 2.1	85.39 ± 7.66	0/10	8/10

Table 4: Monte Carlo summary over 10 randomized environments. ADMM gives the best sparsity-effort profile but does not achieve dynamically feasible collision-free navigation. The difference between X safety and Z safety is the central diagnostic result.

is not safe for the dynamically feasible trajectory. The X trajectory is collision-free in $0/10$ trials, even though the auxiliary trajectory Z is collision-free in $8/10$ trials.

This difference between X safety and Z safety is the most important result of the randomized experiments. The projection step often succeeds at making Z safe, but the actual trajectory X that satisfies the system dynamics still collides. Therefore, the ADMM method should not be interpreted as a safe planner, even though it produces sparse controls and often produces a safe-looking auxiliary trajectory.

Failure Analysis

Why Z Safety Does Not Imply X Safety

The solver struggles to close the consensus gap between Z and X near obstacle boundaries. The obstacle projection is applied to Z, not directly to X. ADMM tries to make $X = Z$, but after a finite number of iterations the two variables may still disagree. Since the dynamics are enforced only on X, the actual feasible trajectory can pass through obstacles even when Z lies on or outside obstacle boundaries. The ADMM method has Z clearance near zero and Z is collision-free, but X has minimum clearance -1.483. The method satisfies the relaxed constraints for sparsity and auxiliary safety, but fails to yield a safe dynamic trajectory.

Why Sparsity and Safety Compete

The $\ell_{2,1}$ penalty encourages long zero-input intervals. To still reach the same goal in the same horizon, the remaining active controls become larger impulses. This is visible in the peak values: the ADMM method has a peak input of 34.79, larger than the quadratic baseline's 23.30. The sparse formulation traded continuous small inputs for brief, high-magnitude impulses. Although the reported effort proxy decreases, the larger peak input suggests that the resulting controls may be less physically realistic without actuator limits.

This also helps explain why stronger sparsity does not improve clearance. The lambda sweep shows that larger λ increases sparsity and decreases the effort proxy, but the dynamic trajectory remains unsafe. The solver achieves sparsity but fails to satisfy obstacle constraints.

Why More Iterations Are Not Enough

The iteration-budget sweep shows that increasing the budget from 50 to 1000 iterations does not make X collision-free. This indicates that early stopping is not the primary cause of the collisions. The combination of greedy projection and the sparsity objective likely creates a local minimum where Z rests on the obstacle boundary while X intersects it.

Limitations and Future Work

The first limitation is the greedy obstacle projection. For multiple obstacles, sequential radial projection is not the exact projection onto the obstacle-free set, and it can behave poorly when obstacles are close together or when dynamics force the trajectory through a narrow region. Future implementations could address this using sequential convex programming directly on the dynamic trajectory. The second limitation is the absence of actuator limits. Without saturation bounds, the sparse controls generate unrealistic peak impulses. Future work should include bounds on acceleration or input magnitude.

The third limitation is the simple double-integrator model. Therefore, the theoretical sparsity achieved does not necessarily translate to physical energy efficiency. Subsequent work should incorporate nonlinear dynamics and receding-horizon control.

The fourth limitation is that safety is checked only at discrete timesteps. A trajectory could still collide between sampled points. Continuous-time collision checking or denser interpolation would be necessary for a stronger safety claim.

Future work should replace the greedy projection with a convexified subproblem. Adding trust regions and actuator constraints would also improve physical viability.

Conclusion

This project studied an ADMM approach for sparsity-promoting trajectory optimization in a 2D multi-obstacle navigation problem. The $\ell_{2,1}$ penalty successfully encouraged temporal sparsity, reducing active control timesteps and lowering the effort proxy relative to an obstacle-blind quadratic baseline. The block soft-thresholding update behaved as expected, and the projection step often produced safe auxiliary trajectories.

However, the experiments show that the current ADMM splitting is not a reliable safe trajectory planner. The core issue is that obstacle projection is applied to the auxiliary variable Z, while the dynamically feasible trajectory is X. In both the fixed environment and randomized trials, Z safety did not transfer to X safety. In conclusion, the approach achieves sparsity but falls short on safety.

These findings highlight a specific failure mode in applying projection-based ADMM to nonconvex trajectory constraints. Future work can enforce obstacle constraints directly on the physically feasible trajectory and incorporate actuator limits.

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